ARTUR KASHPERSKIY

Software Engineer

artkashpers@gmail.com | sm5art.github.io | artur-kashperskiy-9171ab11a | sm5art

EXPERIENCE

Software Engineer
Autodesk

Sept 2022 – Dec 2023 | San Francisco, CA

- Developed infrastructure for user profile platform across Autodesk products, focusing on authentication and authorization at profile.autodesk.com.
- Implemented Observability dashboards and alarms on AWS Cloudwatch and Splunk to monitor latencies, errors, and call counts for user profile API services.
- Optimized infrastructure and internal libraries, reducing API latency from 1.5s to an average of 0.3s, while enhancing API functionality. Implemented rate-limiting for improved stability and performance, and enhanced Redis cache and HTTPS client libraries through rigorous environment testing.
- Rotated on-call support every 8 weeks, utilizing internal tools like Dynatrace, PagerDuty, Splunk, and Cloudwatch for diagnosis and monitoring. Contributed to setting up wikis for run-books on alarming and diagnosis.
- GraphQL, Node.js, Serverless, AWS (SQS, S3, Lambda, DynamoDB, Kinesis, Cloudwatch), Redis, Terraform, Splunk, PagerDuty, Dynatrace.

Software Engineer Intern
Autodesk

June 2021 – September 2021 | San Rafeal, CA

- Worked with User Profile team to build front end for new landing page for when users first hit profile.autodesk.com (to manage a users profile).
- Worked with designers to build a few reusable React components and microfrontend that supports fast loading of the new landing page. Learned JS technologies for localization, GraphQL queries and microfrontend routing.
- GraphQL, React, jenkins, nodeJS, Dynatrace, Adobe AEM, Figma

Software Engineer Intern
Valravn Capital

January 2021 – March 2021 | NY, Remote

- Built up new infrastructure and helped team with moving and consolidating some fundamental macroeconomic data fetching processes for research purpose to new Airflow infrastructure to help reduce costs spent on AWS.
- Built an experimental fintech full stack web application for optimizing and simulating many different portfolios with different exposure options using plotly Dash.
- Apache Airflow, Python, AWS EC2, AWS SageMaker, Plotly Dash

Software Engineer Intern
Heali.ai

July 2019 – October 2019 | Santa Monica, CA

- Built APIs with Express.js and serverless to support feature for mobile app showing diet and nutrition information when scanning food product barcodes with camera.
- Built front end functionality for food product barcode scanning with Swift native OCR plugins in React Native App.
- Added some new infrastructure for bug reporting (Phabricator) and integrated it with a shake to bug report frontend feature to the app.

SKILLS

Python | JavaScript | Java | C | C++ | C# | MATLAB | React | GraphQL | React Nativ | Docker | Kubernetes | Unreal Engine | Unity | MongoDB | AWS EC2 | API Gateway | Lambda | S3 | MariaDB | Redis | Memcached | Terraform

PROJECTS

demix

- Developed a web tool enabling music producers to upload audio files and receive individually separated vocals and instrumental tracks, leveraging machine learning for remixing instead of manual equalization. Launched on AWS EC2 and Docker for 2 months, achieving 8000 weekly active users before shutdown due to high costs.
- python flask server, GatsbyJS frontend, tensorFlow.
- https://github.com/sm5art/demix
- https://github.com/sm5art/demix-frontend

genetic pong

- Enhanced a friend's pong game in Python using the pygame package by implementing a genetic algorithm from scratch. This algorithm utilized an unsupervised fitness heuristic to optimize the performance of the pong "brain". Training took approximately 30 minutes or 30 generations to converge, resulting in the creation of an unbeatable AI player.
- python, numpy, pygame
- YouTube Training Demo
- https://github.com/sm5art/genetic-pong

recipedb

- Built an LLM based application that scrapes popular recipe blog hosts for recipes and gets relevant information such as ingredient/title and helps recommend recipes to the user based on their preferences.
- Used OpenAI ChatGPT APIs for parsing web text for valid recipe information and stored in MongoDB.
- Utilized TFIDF vectorizer models to help find related recipes.
- python, MongoDB, OpenAI GPT, remix/React JS

EDUCATION

B.S. in Applied Physics

- React Native, serverless, nodeJS, Apache Phabricator, AWS, Swift

Software Engineer Intern

Nipun Capital

June 2017 – September 2017 Foster City, CA

- Built up data pipeline for alpha generation and web scraping sources in emerging markets (exchange sites) using python and Apache Airflow (one job required to break old text capchas using CNNs). Moved old alpha generation cronjobs to new Airflow infrastructure.
- Worked on a small NLP research project investigating sentiment analysis of conference call transcripts (financial sentiment dictionary and word2vec methods) and backtested using internal tools.
- python, Google Cloud, Apache Airflow, tensorflow

Software Engineer Intern

Minted

June 2016 – August 2016 San Francisco, CA

- Wrote unit/integration tests improving coverage of marketing insert logic in Minted fulfillment state machine and fixed bugs.
- Built an API endpoint and front end for an analytics dashboard used as a productivity tracking tool for our stationary design auditing team.
- BackboneJS/ReactJS, Python, MySQL, Vagrant, puppeteer

University of Washington

Sept 2017 – December 2022

Notable coursework: Algorithms and Data Structures, Vector Calculus/Complex Analysis, Probability I, Artificial Intelligence, MATLAB for Numerical Analysis, Quantum Mechanics, Digital/Analog Electronics